Quantify outliers

1

a 131 **b** 47 **c** 25, 30 **e** 12

f 21°C

2 (16, 3)

3

a 2009 **b** 2006

4 2002

5

a

i 3 **ii** 2 **iii** 4 **iv** 2

v −1 **vi** 7

b 9

6

a

i 11 **ii** −10.5 **iii** 33.5

b

i 10 **ii** −9 **iii** 31

7

a

i

Minimum	1
Q1	2
Median	4
Q3	6
Maximum	9

ii 4 **iii** −4 **iv** 12

v Yes **vi** Yes

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b

i

Minimum	2
Q1	3
Median	6
Q3	9
Maximum	10

ii 6 **iii** -6 **iv** 18

v No **vi** No

c

i

Minimum	1
Q1	5
Median	7
Q3	10
Maximum	12

ii 5 **iii** -2.5 **iv** 1

v Yes **vi** No

8

a

i

Minimum	1
Q1	2
Median	5
Q3	8
Maximum	10

ii No **iii** No

b

i

Minimum	4
Q1	6
Median	9
Q3	10
Maximum	11

ii Yes **iii** No

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9

a

Minimum	1.1
Lower quartile	1.3
Median	1.7
Upper quartile	2.1
Maximum	2.4

b Yes because the upper fence is 3.3 and her score is greater than this.

10

a

Minimum	71
Lower quartile	73
Median	76
Upper quartile	79
Maximum	84

b No

11 21

12 24

13

a 21, 21, 21, 24, 25, 26, 26, 26, 26, 27, 28, 29, 30, 30, 32, 41, 41, 46, 46, 47, 76

b 28

c 41

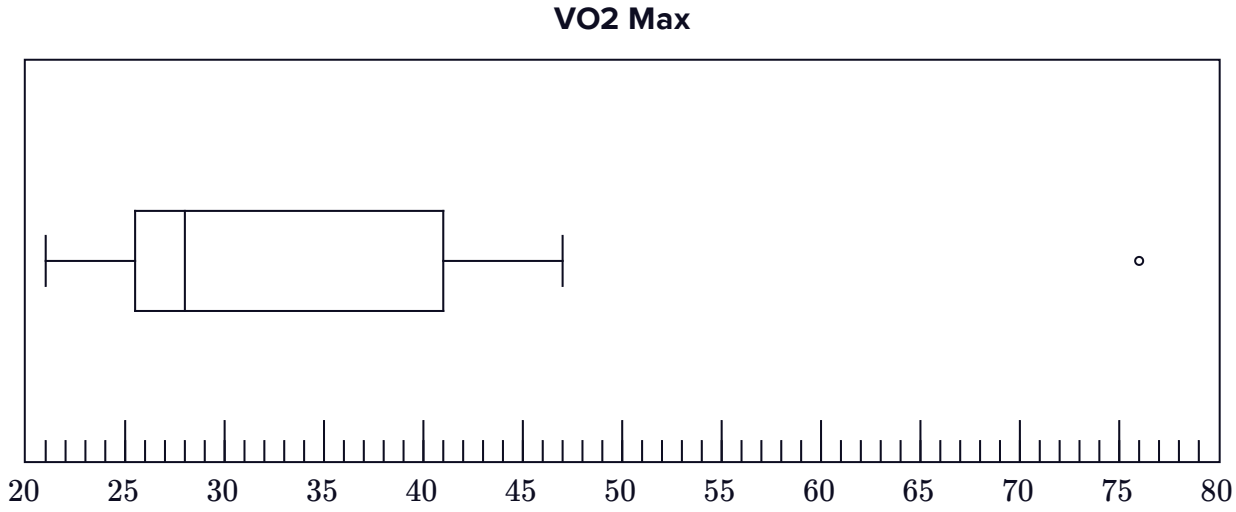
d 25.5

e 23.25

f 76

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g



h The majority of this group of people are likely to do none to moderate amounts of exercise.

Effects of outliers

14

a

i

Mean:	46
Median:	43
Mode:	30
Range:	84

ii 109

iii

Mean:	39
Median:	40
Mode:	30
Range:	28

iv The mean, median and range decreased, but the mode stayed the same.

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b

i

Mean:	1.92
Median:	1.7
Mode:	0.9
Range:	3.8

ii 4.7

iii

Mean:	1.61
Median:	1.5
Mode:	0.9
Range:	1.9

iv The mean, median and range decreased, but the mode stayed the same.

c

i

Mean:	4540
Median:	4750
Mode:	4700
Range:	3800

ii 1500

iii

Mean:	4877.78
Median:	4800
Mode:	4700
Range:	800

iv The mean and median increased, the range decreased, but the mode stayed the same.

15

a Larger **b** Smaller **c** Larger **d** Smaller

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16

a The mode does not change.

b The range always decreases.

c If the outlier removed was the largest value then the mean will decrease. If the outlier removed was the smallest value then the mean will increase.

d If the outlier removed was the largest value then the median may decrease or remain the same. If the outlier removed was the smallest value then the median may increase or remain the same.

17 2

18

a 11 **b** 6

19 44.56

20 18 kg

21 13.3 kg

22

a 11.2 mm **b** 11.15 mm

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